

Approach and Methodology

The objective of this task was to train a supervised machine learning model to predict taxi fares at the trip level. The provided dataset contained several raw features associated with each trip, including trip distance, passenger count, pickup and dropoff information, payment details, and fare-related attributes.

During the data preparation stage, certain features such as `tip_amount`, `tolls_amount`, and `total_amount` were excluded from the training process to prevent data leakage. These variables are typically known only after or near the completion of a trip and are either directly related to the target variable (`fare_amount`) or would not be available at prediction time. Additionally, these columns are not present in `prediction_set.csv`, making their exclusion necessary to ensure consistency between training and production.

Following an exploratory data analysis (EDA) phase, several additional features were engineered to improve the predictive power of the models. These included:

- **Trip Duration** – derived from pickup and dropoff timestamps.
- **Pickup Hour** – to capture potential time-of-day effects.
- **Day of Week** – to account for differences in travel patterns throughout the week.
- **Weekend Flag** – indicating whether a trip occurred on a weekend.
- **Average Speed** – derived from trip distance and duration to capture traffic-related effects and trip characteristics.

The data was also cleaned and transformed into appropriate data types before modeling. Correlation analysis and exploratory analysis were performed to better understand feature relationships and identify the strongest predictors of fare amount. As expected, trip distance and trip duration emerged as the most influential features.

Model Development

Multiple experiments were conducted using both a baseline model and a more advanced machine learning model:

- **Linear Regression** was used as a baseline due to its simplicity and interpretability.
- **XGBoost Regressor** was used as a more flexible model capable of capturing non-linear relationships and feature interactions.

Cross-validation was incorporated during experimentation to obtain a more reliable estimate of model performance and to support hyperparameter tuning. Hyperparameter optimization was performed across multiple configurations to identify the best-performing model settings.

Across the experiments, XGBoost consistently outperformed Linear Regression, suggesting that the relationship between the input features and fare amount contains meaningful non-linear patterns that a linear model could not fully capture. While cross-validation did not lead to

substantial performance improvements, it provided confidence that the selected model generalized well to unseen data.

Experiment Tracking and Results

All experiment artifacts are stored within the `model_exp` directory. Each experiment run generates a separate folder containing:

- Model artifacts
- Evaluation metrics (MAE, MSE, and RMSE)
- Experiment-specific configurations

The final selected XGBoost model was used to generate predictions for the provided prediction dataset. The resulting predictions are available in the repository's root directory with the name `ride_level_predictions.csv`.

Future Improvements

Several enhancements could further improve the scalability and maintainability of this solution:

1. MLflow Integration

Experiment tracking is currently implemented manually. Integrating MLflow would provide a centralized platform for tracking:

- Experiment parameters
- Evaluation metrics
- Model artifacts
- Model versions

This would significantly improve reproducibility and model governance.

2. Containerization

Containerizing the application using Docker would improve portability and deployment consistency. This would allow the solution to run reliably across different environments without requiring manual dependency management or environment configuration.

3. Additional Feature Engineering

Further experimentation could be conducted to identify additional predictive features and evaluate their impact on model performance. Potential areas include location-based features, historical demand patterns, and external contextual information.

4. Larger Training Dataset

Training on a larger and more diverse dataset would likely improve model generalization and robustness. Additional data would allow the model to learn a broader range of travel patterns and reduce sensitivity to dataset-specific variations.